

DNA Layer Inherits Paper 1's Orchestration Rules and Behavior Substrates

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This work formalizes an inheritance edge between Paper 2 (Li, April 2026) and Paper 1 (Li, April 2026) of the Coordination Knowledge Subset (CKS) theory series. It does not introduce new axioms. Its sole contribution is to establish, in precise operational form, that Paper 2's DNA layer is a strict extension ($\supset$) of Paper 1's orchestration rules and behavior substrates—carrying every governance commitment those objects carry under Paper 1, while adding naming, versioning, directed selection as the governed improvement mechanism, and two new named object types.

Abstract

Paper 1 of the Coordination Knowledge Substrate (CKS) series establishes orchestration rules and behavior substrates as the primary objects through which human governance specifies LLM behavior. Both are substrate content: human-governed, stored in the substrate, accessible to the LLM under the mediator role, subject to conflict preservation, and modifiable at any time through the three rights. Paper 2 names the cell's governance layer explicitly as the **DNA layer**—a unified, bounded layer within the cell containing all specifications that govern the cell's behavior. This note formalizes the inheritance relationship: the DNA layer strictly contains ($\supset$) Paper 1's orchestration rules and behavior substrates. Every governance commitment Paper 1 makes for those objects travels unchanged into the DNA layer. What Paper 2 adds is not a new governance framework but four structural additions: explicit layer naming and bounding, a versioning commitment through the lineage chain, directed selection as the named governed mechanism for DNA improvement, and two new named object types (harness substrates, verification substrates) within the broader category of behavior substrates. These additions explain why the relationship is $\supset$ rather than $\equiv$. The note states the preserved commitments, characterizes the additions, explains the $\supset / \equiv$ distinction, identifies three prior-art claims the inheritance edge forecloses, and provides an operational test.

1. Why the inheritance edge needs to be stated

The CKS theory series is constructed on a principle of cumulative inheritance: later papers extend earlier commitments without redefending them. Paper 2 explicitly states that all Paper 1 architectural commitments hold at every level of its three-level structure (cell, aspect, Self), but it does not separately analyze which Paper 2 architectural objects map to which Paper 1 architectural objects. That mapping is

implicit in the papers' design but is not formalized as a named inheritance edge.

Formalizing the edge matters for two reasons. The first is prior-art clarity. An adversarial reader of Paper 2 who encounters the DNA layer described as a unified governed layer with versioning, directed selection, and multiple named object types might characterize it as a novel governance structure unrelated to Paper 1's orchestration rules—a new architectural commitment requiring fresh defense. This characterization is incorrect, and this note establishes publicly that it is incorrect: the DNA layer is the orchestration rules and behavior substrates from Paper 1, named, bounded, and extended. The second reason is downstream work. Papers 3 and beyond treat DNA-layer content as a standard object in inter-Self exchange. Every property of that content's governance follows from Paper 1, not from fresh commitments in Papers 2 or 3. Stating the inheritance edge makes the derivation path for downstream governance properties explicit.

2. Paper 1's commitments for orchestration rules and behavior substrates

Paper 1 establishes orchestration rules as the human-authored specifications that determine cell-level behavior: what the LLM is authorized to do on the substrate's behalf, how conflicts are handled, what output counts as valid, what conditions trigger which response. Paper 1 also commits to behavior substrates—the broader category of substrate objects that specify how the cell should operate in different contexts. Both are governed under the same framework:

■ **Substrate content.** Orchestration rules and behavior substrates are substrate content: structured representations carried in the persistent substrate, not in the LLM's weights or in runtime middleware.

■ **Human-governed (three rights).** The three rights—to inspect, to modify, to override—apply fully to orchestration rules and behavior substrates. These rights are architectural properties, not procedural promises.

■ **AI-as-mediator.** The LLM reads from orchestration rules and behavior substrates as its behavioral specification. It does not write to them outside human authority. No shadow behavioral specifications exist.

■ **Source of truth.** The substrate's orchestration rules and behavior substrates are the authoritative record of how the cell is specified to behave. There is no other authoritative location.

■ **Conflict preservation.** Contradictions in the governance layer are preserved as first-class addressable objects rather than silently resolved, with provenance attached.

■ **Path retraceability.** Changes to orchestration rules and behavior substrates carry provenance metadata sufficient for an observer to reconstruct the causal path of any modification.

These six commitments constitute the governance framework Paper 1 applies to its orchestration rules and behavior substrates. No Paper 2 architectural commitment reduces, replaces, or supersedes any of them.

3. The inheritance: what is preserved

The DNA layer is the named, bounded, and versioned consolidation of Paper 1’s orchestration rules and behavior substrates within Paper 2’s two-layer cell architecture. Every piece of content in the DNA layer is, from a governance standpoint, subject to exactly the six commitments enumerated in §2:

■**Substrate content identity is preserved.** DNA-layer content is substrate content. It resides in the persistent substrate, not in the instinct layer or in any component outside the substrate. The DNA layer is a named portion of the substrate the cell owns, carrying the same substrate-content properties it has always carried under Paper 1.

■**Three rights are fully preserved.** Any DNA-layer content—orchestration rules, harness substrates, verification substrates, coordination rules, purpose statements, content-domain specifications, level determination records—falls within the three rights. LLM drafting of DNA-layer content is permissible under human authority; LLM-committed DNA content outside human authority is not admissible, by direct inheritance from Paper 1 §3.3’s authority-vs-labor distinction.

■**AI-as-mediator role is preserved.** The LLM reads from DNA-layer content as its behavioral specification under the same mediator role Paper 1 establishes. No shadow orchestration rules exist outside DNA-layer substrate content.

■**Source-of-truth property is preserved.** The DNA layer is the authoritative record of the cell’s behavioral specification. There is no other location where the cell’s behavioral specification is canonical.

■**Conflict preservation is preserved.** Contradictions within DNA-layer content become persistent first-class substrate objects with provenance, not silently resolved.

■**Path retraceability is preserved.** Every modification to DNA-layer content carries provenance metadata sufficient for reconstruction. This applies to routine modifications through the three rights and to governed changes through directed selection.

4. What Paper 2 adds

The DNA layer is a strict extension, not merely a renaming. Four structural additions distinguish the DNA layer from Paper 1’s unstructured set of orchestration rules and behavior substrates:

4.1 Explicit naming and layer bounding

Paper 1 establishes orchestration rules and behavior substrates as substrate content but does not name them collectively as a layer or distinguish them structurally from operational records in the substrate. Paper 2 introduces the explicit architecture of two layers within every cell—DNA and Action—and names the DNA layer as the bounded container of all behavioral-specification content. This naming makes the governance layer of a cell a first-class architectural object: observable, separable from the action layer, and addressable as a whole for mating, versioning, and directed selection. The governance

content of what the layer holds travels from Paper 1 unchanged.

4.2 DNA versioning via the lineage chain

Paper 1 commits to the modify right—humans may change orchestration rules at any time—but does not commit to a versioning structure that records each governed change as a version event in an explicit lineage chain. Paper 2 adds this commitment: the DNA layer carries an explicit version history, with each directed selection event recorded in the lineage chain. The version history is substrate content and therefore itself human-governed. An observer of a Paper 2 deployment can trace the DNA layer’s governance history from the current state back through every governed change, not merely inspect the current state. This is the single most consequential structural addition, because it gives the governance layer a temporal dimension that Paper 1’s modification commitment does not provide.

4.3 Directed selection as the named governed improvement mechanism

Paper 1 allows modification of orchestration rules and behavior substrates through the modify right but does not name a specific mechanism for governed improvement—how humans deliberately refine the cell’s behavioral specification toward governance-defined goals. Paper 2 commits to directed selection as that mechanism: a deliberate, goal-oriented process that selects DNA-layer content by intent and outcome together, producing a predictable governance trajectory. Directed selection is governed through the standard Paper 1 authority architecture (the three rights apply to the selection criteria themselves, which are substrate content), but the mechanism—purposeful directed improvement of a versioned behavioral-specification layer—is new. It distinguishes DNA evolution from instinct evolution (undirected mutation) and from action-feedback evolution (the action-to-DNA loop through operational experience).

4.4 Harness substrates and verification substrates as named object types

Paper 1 introduces behavior substrates as the broad category of substrate objects specifying how the cell operates in different contexts. Paper 2 names two specific object types within this category:

■ *Harness substrates* are behavior substrates that specify which sub-substrates are active for current cell activity—governed per-deployment activation under Paper 1’s authority architecture. They are substrate content, fully subject to the three rights.

■ *Verification substrates* are behavior substrates that specify parallel-run behavior: running instinct-layer and reasoning-layer operations simultaneously to detect divergence and govern integration. They are substrate content, fully subject to the three rights.

Both additions are object-type additions within the DNA layer, not new governance principles. An implementer who knows Paper 1’s behavior-substrate governance commitments applies exactly the same governance framework to harness and verification substrates: inspect, modify, override as needed, with no additional requirements from the fact that the object is specifically named rather than an unnamed behavior substrate.

5. Why $\supset$ rather than $\equiv$

A strict extension ($\supset$) differs from an identity ($\equiv$) in that the extending object satisfies everything the extended object commits to and adds further commitments. The DNA layer satisfies every governance commitment Paper 1 makes for orchestration rules and behavior substrates—established in §3—and adds the four structural elements described in §4. Because of those additions, the relationship is strict extension rather than identity.

The contrast with C1.05 (the cell $\equiv$ relation) is instructive. Paper 2's cell is identical to Paper 1's cell in architectural definition: the cell is the atomic unit, all Paper 1 commitments apply at the cell level, and Paper 2 adds no new structural elements to the cell itself. The DNA layer is not an identical preservation; it is a named, versioned, directed-selection-governed, two-new-object-type extension of what Paper 1 called orchestration rules and behavior substrates. Hence $\supset$.

The $\supset$ direction forecloses a specific misreading: that the DNA layer's additions (versioning, directed selection, harness substrates, verification substrates) are independent inventions not derived from Paper 1. The $\supset$ relation says the opposite: those additions are built on top of Paper 1's governance framework, not alongside it. Directed selection modifies DNA-layer content, which is substrate content under the three rights; without Paper 1's governance framework, directed selection is undefined as a governed mechanism. Harness and verification substrates are governed as behavior substrates under Paper 1's commitments; without Paper 1's behavior-substrate governance, they have no governance commitments to inherit.

6. Prior-art significance

Three adversarial claims are foreclosed by this inheritance edge:

■**Claim (a):** *Paper 2's DNA layer introduces novel governance objects unrelated to Paper 1's orchestration rules.* Foreclosed. The DNA layer is the named, bounded consolidation of Paper 1's orchestration rules and behavior substrates. No DNA-layer governance object is independent of Paper 1's orchestration-rule and behavior-substrate category.

■**Claim (b):** *DNA versioning is novel relative to Paper 1's modifiable substrate content.* Foreclosed in the form that Paper 2 introduces versioning into a previously unversioned governance context. Paper 1 commits to the modify right for orchestration rules and behavior substrates, establishing that such content is mutable substrate content subject to human authority. Paper 2's explicit lineage chain and version-history commitment is a structured extension of the modify right's implicit record-keeping. The versioning adds structure and explicit commitment; it does not introduce mutability into a previously immutable governance layer.

■**Claim (c):** *Harness substrates and verification substrates are novel inventions unrelated to Paper 1's behavior substrates.* Foreclosed. Both are named species of behavior substrate under Paper 1's framework. The governance commitments that apply to them are Paper 1's behavior-substrate governance commitments applied to these two specific object types. The novelty in Paper 2 is the naming and specification of what each object type carries, not the introduction of

governance commitments for them.

7. Relationship to C1.09

This note covers the DNA layer as the inheritance of Paper 1's orchestration rules and behavior substrates. C1.09 will cover the corresponding inheritance edge for the action layer: the Paper 2 action layer (recorded task instances and their outputs) as the inheritance of Paper 1's substrate content for operational records. The DNA/action distinction is Paper 2's within-cell structural addition; together C1.08 and C1.09 establish that both layers of every Paper 2 cell are extensions of Paper 1 substrate content, each inheriting appropriate Paper 1 governance commitments.

8. Operational test

An observer examining a Paper 2 deployment can apply the following test to verify that Paper 1's orchestration-rule governance commitments hold for DNA-layer content, independently of the DNA versioning structure:

- (1) Take any DNA-layer content in the deployment—a coordination rule, a purpose statement, a harness substrate, a verification substrate, any element within the DNA layer.
- (2) Verify that a human with appropriate access can read that content directly, in inspectable form, without scheduling, approval, or LLM intermediation as a precondition. If this fails, the inspect right is not preserved for DNA-layer content, and Paper 1's governance commitment is not satisfied by this deployment regardless of what the versioning records show.
- (3) Verify that a human with appropriate access can modify that content with the change taking effect as substrate state. If this fails, the modify right is not preserved for DNA-layer content.
- (4) Verify that no LLM operation, vendor policy, or runtime middleware layer can in principle prevent (2) or (3). If this fails, the architectural character of the three rights is not preserved.
- (5) Verify that any DNA-layer content authored by an LLM was subject to human authority before it took effect—that LLM-drafted content was placed in the DNA layer under governance authorization, not committed autonomously. If this fails, Paper 1 §3.3's authority-vs-labor distinction is not preserved.
- (6) Verify that contradictions within DNA-layer content are preserved as first-class addressable substrate objects with provenance, not silently resolved. If this fails, conflict preservation is not preserved for DNA-layer content.

A deployment that passes all six checks satisfies Paper 1's governance commitments for the DNA layer independently of its versioning structure. The versioning structure (the lineage chain recording directed selection events) is then verified separately as a Paper 2 addition on top of the passing Paper 1 baseline. A deployment that fails any of (2)–(6) does not implement the DNA layer as the inheritance of Paper 1's orchestration rules and behavior substrates; it implements something structurally resembling the DNA layer but without the governance commitments the inheritance edge requires.

9. Conclusion

Paper 2's DNA layer strictly extends ($\supset$) Paper 1's orchestration rules and behavior substrates. Every governance commitment Paper 1 establishes for those objects—substrate content identity, three rights, AI-as-mediator, source-of-truth, conflict preservation, path retraceability—travels unchanged into the DNA layer. What Paper 2 adds is four structural elements: explicit layer naming and bounding that makes the governance layer of a cell a first-class architectural object; an explicit version history through the lineage chain that records each directed selection event; directed selection as the named, goal-oriented governed mechanism for DNA improvement; and two new named object types (harness substrates, verification substrates) within the broader category of behavior substrates. These additions are built on Paper 1's governance framework, not alongside it.

Downstream work that operates on DNA-layer content—whether computing mating operations, tracing lineage chains, executing directed selection, or exchanging DNA-layer content across the inter-Self perimeter in Paper 3—inherits Paper 1's governance commitments for that content without fresh defense. The governance properties of DNA-layer content in any CKS-governed deployment are Paper 1's properties.

Source papers

Li, W. (2026a). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026b). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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